

Study Report: Multi-Agent Systems - Agentic AI Engineer

Written by Gagan Pasupuleti

Family: Agentic AI Engineer
Level: Advanced
Chapters: 7
Source: Reference: Multi-Agent Systems - Gerhard Weiss

Official sources and free libraries

- <https://mitpress.mit.edu/9780262737317/multiagent-systems/>

Summary

Study report for **Multi-Agent Systems** by Gerhard Weiss (Advanced level) mapped to the Agentic AI Engineer role. Theory and design of coordinated autonomous agent systems.

Chapters

Track: Book-Intro

Chapter 1: About Multi-Agent Systems [Advanced]

Chapter focus: About Multi-Agent Systems** **(Level: Advanced)**

This study report summarizes **Multi-Agent Systems** by Gerhard Weiss for the Agentic AI Engineer role. The resource is rated Advanced level. Theory and design of coordinated autonomous agent systems. Use this report alongside the official material to prepare for CodeQuest review.

Code Reference:

```
# Multi-Agent Systems
# Author: Gerhard Weiss
# Role: Agentic AI Engineer
# Level: Advanced
```

What it shows: Metadata block records the source book and target role family.

Topic guide

What it actually is

A structured study report based on Multi-Agent Systems.

When to use it

When learning Agentic AI Engineer skills at Advanced level.

Where to use it

Agentic AI Engineer training paths and certification prep.

Real use example

A learner reads this report before starting the full book or free online guide.

Track: Book-Context

Chapter 2: Why This Book Matters for Your Role [Advanced]

Chapter focus: Why This Book Matters for Your Role *(Level: Advanced)**

Agentic AI Engineer professionals use ideas from Multi-Agent Systems to solve real workplace problems. Theory and design of coordinated autonomous agent systems. This chapter explains how the book fits into your learning path and what you should be able to do after studying it.

Code Reference:

Role: Agentic AI Engineer

Book focus: Theory and design of coordinated autonomous agent systems.

Recommended level: Advanced

What it shows: Connects the book topic to the job role outcomes.

Topic guide

What it actually is

Role-aligned learning connects theory to job tasks.

When to use it

During career planning and syllabus design.

Where to use it

Agentic AI Engineer bootcamps and CodeQuest teacher assignments.

Real use example

A teacher assigns this book report before a advanced skills checkpoint.

Track: Book-Topics

Chapter 3: Key Topics Covered [Advanced]

Chapter focus: Key Topics Covered *(Level: Advanced)**

The main topics in Multi-Agent Systems include practical concepts described as: Theory and design of coordinated autonomous agent systems. Study each topic with hands-on practice. Take notes on definitions, workflows, and examples that match tools used in Agentic AI Engineer jobs today.

Code Reference:

- Topic 1: core concept
- Topic 2: core concept
- Topic 3: core concept
- Topic 4: core concept

What it shows: Topic list guides what to study chapter-by-chapter in the source.

Topic guide**What it actually is**

Topic maps turn a book into actionable learning objectives.

When to use it

Before reading and while building chapter summaries.

Where to use it

Self-study, flipped classroom, and revision.

Real use example

A student maps each book chapter to a CodeQuest report section.

Track: Book-Applied

Chapter 4: Applied: Long-Horizon Agent Workflows [Advanced]**Chapter focus: Long-Horizon Agent Workflows** *(Level: Advanced)**

Tasks spanning hours need job queues, idempotent tools, progress UI, and cancellation. Break into milestones; persist artifacts to object storage between steps.

Code Reference:

```
class JobState(TypedDict):
    milestone: int
    artifacts: list[str]
    status: Literal['running', 'paused', 'done', 'failed']
```

What it shows: Typed state documents milestone progress for supervisor and UI.

Topic guide**What it actually is**

Long-horizon patterns scale agents beyond chat session length.

When to use it

Report generation, migration planning, and large code refactors.

Where to use it

Enterprise automation replacing multi-day manual processes.

Real use example

Agent writes section drafts to S3 each milestone - user reviews async.

Chapter 5: Applied: Advanced MCP: Resources and Prompts [Advanced]

Chapter focus: Advanced MCP: Resources and Prompts** *(Level: Advanced)

MCP servers can expose resources (file URIs) and prompt templates besides tools. Hosts prefetch resources into context - design least-privilege resource scopes.

Code Reference:

```
{
  "method": "resources/read",
  "params": {"uri": "file:///policy/handbook.md"}
}
```

What it shows: resources/read returns content host may inject before agent acts.

Topic guide

What it actually is

Full MCP surface enables richer integrations than tools alone.

When to use it

IDE and ops agents needing structured context injection.

Where to use it

Policy handbooks, runbooks, and schema catalogs.

Real use example

Agent pulls latest on-call runbook resource URI - always current version.

Chapter 6: Applied: Evaluating Agent Reliability [Advanced]

Chapter focus: Evaluating Agent Reliability** *(Level: Advanced)

Define task success criteria per scenario (tool called, correct arg, final answer contains X). Run nightly regression; track success rate, steps-to-complete, cost per task.

Code Reference:

```
def eval_task(run):
    assert run.tool_calls[0].name == 'search'
```

```
assert 'Q3 revenue' in run.final_answer
assert run.steps <= 8
```

What it shows: Assert-style evals codify pass/fail for CI agent regression.

Topic guide

What it actually is

Agent evals measure reliability beyond single-turn LLM benchmarks.

When to use it

Before promoting agent prompt or tool changes.

Where to use it

Platform teams and regulated agent deployments.

Real use example

Success rate drops from 92% to 81% after model swap - rollback triggered.

Track: Book-Plan

Chapter 7: Study Plan and Practice [Advanced]

Chapter focus: Study Plan and Practice** *(Level: Advanced)

Finish Multi-Agent Systems with a weekly plan: read one section, write a summary, complete one exercise, and reflect on how it applies to Agentic AI Engineer. If a free edition exists, practice every example. Submit your notes and one mini-project demo for teacher review.

Code Reference:

```
Week 1: Read + notes
Week 2: Exercises
Week 3: Mini project
Week 4: Review + quiz
```

What it shows: A 4-week plan turns reading into demonstrable skill.

Topic guide

What it actually is

Structured study plans improve retention and portfolio outcomes.

When to use it

After finishing the key topics chapters.

Where to use it

CodeQuest end-of-unit assessments.

Real use example

A learner completes the study plan and uploads a advanced project screenshot.